

Client misses inventory updates while player is manipulating items - causes invisible items

MC-29318 Networking

Status	Resolved / Fixed
Affected	Minecraft 1.6.2, Minecraft 14w30c, Minecraft 1.8.1, Minecraft 1.8.2-pre6, Minecraft 15w44b, Minecraft 1.8.9 (+43 more)
Fixed in	21w10a
Reported	2013-08-24

STEPS TO REPRODUCE

1. Create a Creative world with default settings.
2. Place a furnace.
3. Obtain charcoal or coal, a raw steak, and another item for the inventory.
4. Open the furnace GUI and place the charcoal or coal in the fuel slot.
5. Place the raw steak in the input slot and keep the other item in the inventory.
6. Wait until the furnace progress bar is almost at the end.
7. Observe that the progress bar is nearly complete.
8. Continuously left-click the other item in the inventory while the furnace finishes cooking.
9. Observe that the progress bar has finished but the output slot appears empty; click the output slot and observe that the cooked item is received.

OBSERVED BEHAVIOUR

Actual behavior

While the player is repeatedly clicking or moving items in an open container GUI, the client can miss server-side slot updates.

- In a furnace, the server completes the smelting operation and updates the input and output slots, but the client may continue displaying the old input item and an empty output slot.
- The progress bar can indicate that smelting has completed even though the cooked item is not visible in the output slot.
- Clicking the apparently empty output slot still retrieves the cooked item, confirming that the item exists on the server and that only the client display is stale.
- With `/replaceitem`, the server continues replacing the targeted inventory slot, but the client can display that slot as empty while the player is moving the newly inserted items.
- The missing item becomes visible after the relevant inventory interaction ends, the GUI is reopened, or the world is re-entered.

The issue affects the client-side representation of container and player-inventory slots; the server inventory remains correct.

EXPECTED BEHAVIOUR

While the player is moving items with a container or inventory interface open, all server-side inventory changes should be displayed immediately and accurately on the client. Furnace contents and progress should update normally, with the completed item appearing in the output slot without requiring a click. Items inserted or replaced by commands should also remain visible in the correct slots, with the displayed inventory matching the actual contents when the interface is reopened.

ENVIRONMENT

Minecraft Java Edition Versions: 1.6.2, 14w30c, 1.8.1, 1.8.2-pre6, 15w44b, 1.8.9, 16w05b, 1.9 Pre-Release 4, 1.9, 1.9.1 Pre-Release 1, 1.9.1 Pre-Release 3, 1.9.4, 1.10.2, 16w35a, 16w36a, 16w39c, 1.12, 1.12.2, 18w14b, 1.13-pre2, 1.13-pre5, 1.13-pre6, 1.13-pre8, 1.13, 1.13.1, 1.13.2, 18w48a, 18w48b, 18w49a, 1.14 Pre-Release 1,

1.14, 1.14.1 Pre-Release 2, 1.14.2, 1.14.3, 1.14.4, 19w35a, 19w37a, 1.15.1, 1.15.2, 20w07a, 20w17a, 20w18a, 20w20b, 20w21a, 20w30a, 1.16.2, 1.16.3, 20w46a, 20w51a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

When using whatever container that automatically updates the contents of a slot (like a furnace) and if the player, with the container GUI open, is constantly moving items into the inventory, the player won't see the container slot update (like a raw steak being cooked).

How to reproduce

- Open the furnace gui
- Place charcoal/coal and keep some item in your inventory
- Place something to be cooked into the furnace
- When the progress bar is almost at the end, start continuously left clicking an item in the inventory

What should happen is that the item to be cooked doesn't update (if it doesn't happen the first time, just retry), the progress bar has finished, and you won't have any item in the output slot, but if you click in it you will get the item cooked.

This also occurs with the `/replaceitem` command:

How to reproduce

- Put a command block on a clock, with a command such as:

```
/replaceitem entity @p inventory.0 golden_apple
```

→ This will put a golden apple in the first slot of your inventory

- Open your inventory, and move the golden apple to another slot
- As soon as a new one appears, grab it and move it to another slot as well

→ If you got the timing right, the slot will appear blank, even though the command block output in the chat window continues to say "[@: Replaced slot 9 with 1 x [Golden Apple]]"

- Turn off the clock
 - Exit the world
 - Re-enter the world, and open your inventory
- There will be a golden apple in the first slot

Why this happens:

This happens because there's a boolean in `EntityPlayerMP` that's set to true when the server receives a window click packet and then the function that should send the slot updates is called, though if this boolean is true it will not send the packet to the client to update it (while the server-side thinks from the next update on that the client has received the update).

Basically if the server code updates the content of some slots in the container (raw food to be removed, output slot set to cooked food) while the player is moving other items in the inventory.. it won't get the update.

So one way to fix this would be to know which slot is getting modified

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-29318>